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Kidney function in minority children and adolescents with metabolically healthy and unhealthy obesity

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Summary

Metabolic syndrome and/or body mass index (BMI) ≥ 40 kg/m² are risk factors for kidney function decline in the general population. To compare creatinine (Cr), estimated glomerular filtration rate (eGFR) and blood urea nitrogen (BUN) between minority children and adolescents with metabolically healthy obesity (MHO) and metabolically unhealthy obesity (MUO), a chart review was conducted on subjects aged 4 to 20 years with BMI $\geq 95^{\text{th}}$ percentile from July 2014 to April 2017. They were stratified into MHO and MUO groups. Cr, eGFR and BUN were studied. Total $n = 277$: MHO $n = 105$ vs MUO $n = 172$. Cr was higher and BUN was lower in MUO whereas eGFR did not differ between the groups. Using general linear model, we found that metabolic status predicted BUN ($P = .009$) but not Cr or eGFR. When age, sex and Tanner stage matched, BUN, Cr and eGFR were similar between the groups. Higher BUN in MHO could be due to higher dietary protein intake. Subjects were divided into BMI ≥ 40 vs < 40 kg/m², BUN and eGFR were not different. A trend towards higher Cr in those with BMI ≥ 40 kg/m² ($P = .054$) was found; the group being older and taller. After age and height matching, all outcomes were not different. Higher Cr was found in those with elevated blood pressures vs the MHO ($P = .047$). Those with diastolic blood pressure (DBP) $\geq 90^{\text{th}}$ percentile had higher Cr than those with systolic blood pressure $\geq 90^{\text{th}}$ percentile ($P = .017$). Children and adolescents with MUO, and those with BMI ≥ 40 kg/m² did not appear to have early diminished kidney function. Higher Cr, although in normal range, occurred in those with abnormal DBP.

KEYWORDS

creatinine, glomerular hyperfiltration, kidney function, metabolically unhealthy, obesity in children

1 | INTRODUCTION

Obesity in children has a prevalence of 107.7 million children worldwide¹ and is associated with increased risk of development of diabetes, cardiovascular morbidity² and chronic kidney disease (CKD).³ Depending on cardiometabolic risk factors, obesity has been categorized into metabolically healthy or unhealthy. International Diabetes Federation defines metabolic syndrome in adolescents from 10 to 16 years by the presence

of central obesity (waist circumference $\geq 90^{\text{th}}$ percentile) and two other criteria: triglycerides (TG) ≥ 1.69 mmol/L (≥ 150 mg/dL), high-density lipoprotein (HDL) < 1.03 mmol/L (< 40 mg/dL), systolic blood pressure (SBP) ≥ 130 mm Hg and/or diastolic blood pressure (DBP) ≥ 85 mm Hg, fasting plasma glucose (FPG) ≥ 5.55 mmol/L (≥ 100 mg/dL) or type 2 diabetes.⁴ There is no established consensus on the definition of metabolically healthy obesity (MHO) and metabolically unhealthy obesity (MUO) in children.

CKD in adults has been associated with increasing body mass index (BMI) independent of the metabolic health and/or with metabolic syndrome.^{5,6} A 9-year Atherosclerosis Risk in Communities study of 10 096 non-diabetic adults showed that metabolic syndrome was independently associated with CKD after adjusting for hypertension and diabetes.⁵ The Framingham Heart study on obesity sub-phenotypes reported a small but significant increased risk of CKD in adults with MUO compared to MHO, although when analysing stable MHO and MUO phenotypes that did not change to another phenotype overtime, CKD risk did not differ between the groups.⁷ A longitudinal study in 41 194 Korean adults showed that metabolic health status was not protective of CKD. A higher incidence of CKD was seen in MHO and metabolically unhealthy with normal BMI group compared with metabolically healthy with normal BMI group, with the highest risk of CKD seen in the MUO group.⁸ Others found that BMI ≥ 40 kg/m², regardless of metabolic status, had a significant association with CKD and end stage kidney disease (ESKD).⁶

In children, even though the evidence is not as robust, there are several studies showing a positive association of obesity with the risk of development of CKD. In a nationwide population study in Israel on adolescents aged 17 years, a positive association of higher BMI and obesity in adolescents with non-diabetic ESKD was found.⁹ A cross-sectional analysis of 242 adolescents with obesity, aged 13 to 19 years, who underwent bariatric surgery found 17% had proteinuria and 3% had estimated glomerular filtration rate (eGFR) below 60 mL/min/1.73 m².¹⁰ Case-control study done on 113 adolescents with obesity in Iran showed that those with obesity and metabolic syndrome had lower GFRs compared to those without metabolic syndrome.¹¹ The development of CKD associated with obesity is termed as obesity-related glomerulopathy (ORG). ORG entails a spectrum of glomerular disease, ranging from hyperfiltration and glomerulomegaly at one end to focal segmental glomerulosclerosis and fibrosis of the kidney leading to ESKD at the most severe end.¹⁰ The pathogenesis of ORG is multifactorial. The exact mechanisms are unclear and may include (Figure 1)¹²:

- Altered renal haemodynamics: Glomerular hyperfiltration and increased activation of renin angiotensin system leading to increased sodium retention and hypertension.¹³ Glomerular

hyperfiltration is most commonly defined as ≥ 135 mL/min/1.73 m².^{10,14}

- Activation of pro-inflammatory adipokines including leptin stimulates glomerular cell hypertrophy and mesangial cell proliferation¹⁵;
- Anti-inflammatory adipokines including adiponectin have been shown to protect against free fatty acid (FFA) accumulation and metabolic syndrome. Decrease in adiponectin leads to proteinuria, podocyte dysfunction and interstitial fibrosis.¹⁶ Renal lipotoxicity is secondary to accumulation of increased albumin bound FFA causing injury to the interstitial tubules.¹⁷

With this evidence regarding risk of kidney function decline in children and adolescents with obesity, our aim is to study the effect of obesity and metabolic health on kidney function assessed by creatinine (Cr), blood urea nitrogen (BUN) and eGFR. We hypothesize that children and adolescents with MHO have more favourable kidney function than those with MUO.

2 | METHODS

A retrospective chart review was conducted on subjects followed in the paediatric weight management and pediatric endocrinology clinics from July 2014 to April 2017. Subjects were children and adolescents from the age of 4 to 20 years with obesity defined as BMI ≥ 95 th percentile for age and sex. The sample was representative of the study population.

2.1 | Data collection

Three hundred and twenty-five charts were reviewed. The data obtained from the record included: medical history, family history, age, sex, ethnicity, demographics, vital signs, weight, height and BMI, and sexual maturity rating according to Tanner staging. SBP or DBP ≥ 90 th percentile was considered significant based on the national 2004 "Fourth Report on the Diagnosis, Evaluation, and Treatment of High Blood Pressure in Children and Adolescents."¹⁸ Laboratory data within 6 months of the patient visit included: fasting glucose level if available, A1C, comprehensive metabolic panel that included BUN, Cr, and lipid profile. eGFR was calculated using modified Schwartz formula (mL/min/1.73 m²).¹⁹

2.1.1 | Definition of MUO and MHO subtypes

Subjects were classified as MUO when at least one of the following criteria was met:

- A1C $\geq 5.7\%$ or FPG ≥ 5.55 mmol/L (≥ 100 mg/dL)⁴;
- TG ≥ 1.69 mmol/L (≥ 150 mg/dL)⁴;
- HDL < 1.03 mmol/L (< 40 mg/dL)⁴;
- SBP or DBP ≥ 90 th percentile for age, sex and height.¹⁸

Those that did not meet any of the criteria above were classified as MHO. The subjects removed from the data analysis included those with any underlying illnesses affecting the kidneys, that is, history of

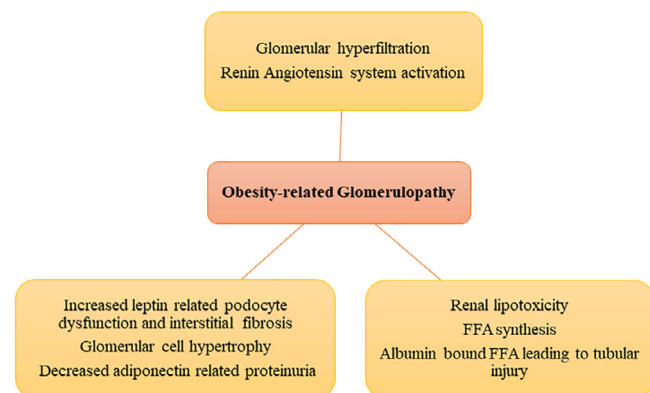


FIGURE 1 Pathogenesis of obesity-related glomerulopathy

urinary tract infections, vesicoureteric reflux disease, hydronephrosis, nephritic syndrome or cystic kidney disease; missing clinical and laboratory information needed for classification; or definition of obesity not met. A total of 277 subjects met the criteria for classification into MUO or MHO and were included in the final analysis.

2.2 | Statistical analysis

t tests and chi-square tests were used. A general linear model was used to assess the relationship between the metabolic health status and the three outcomes: Cr, eGFR and BUN. Data were expressed as mean $\pm$ SD unless otherwise noted. *P*-value of $<.05$ was considered statistically significant.

Additionally, the outcomes of subgroups of MUO based on each variable defining metabolic health status including A1C, TG, HDL and SBP or DBP were compared with the MHO group. Subjects were also divided into BMI ≥ 40 and <40 kg/m² and their outcomes were compared.

3 | RESULTS

A total of 277 subjects, aged 4 to 20 years were stratified into MUO (*n* = 172, 62%) or MHO (*n* = 105, 38%). More than 95% of our subjects were of African-American or Caribbean descent.

3.1 | Characteristics of the MHO and MUO groups

The characteristics of the two groups were compared. Subjects in MUO group were older and had a higher BMI than those in MHO group. Mean age was 12.93 ± 3.29 years (*P* = .012) in MUO vs 11.89 ± 3.31 years in MHO. Mean BMI was 33.64 ± 6.74 kg/m² vs 31.06 ± 5.99 kg/m² (*P* = .001). Tanner staging was more advanced in the MUO (*P* = .058). There were more females in MHO group (*P* = .026). As expected, based on the classification of the two groups; SBP, DBP, A1C, TG were higher and HDL was lower in MUO (Table 1).

3.2 | Predicting kidney function outcomes using metabolic health status

BUN was lower (*P* = .023) and Cr was higher (*P* = .027) in the MUO group. There was no difference in the eGFR between the two groups.

When a general linear model was used to predict the outcomes based on the metabolic health status, metabolic health status predicted BUN (*P* = .009) but not Cr (*P* = .334) or eGFR (*P* = .232).

After matching the groups for age, sex and Tanner stage, the BMI between both groups was similar and there was no difference found in Cr (*P* = .478) and eGFR (*P* = .407), although there was a trend towards higher BUN in the children with MHO (*P* = .052). Furthermore, subgroups of MUO based on each variable affected were compared to the MHO group:

1. Effect of prediabetes on kidney function outcomes

When comparing MUO group with A1C $\geq 5.7\%$ (*n* = 85) against

MHO (*n* = 105), the MUO with A1C $\geq 5.7\%$, were older (13.61 ± 2.81 vs 11.89 ± 3.31 years, *P* < .001), had a higher BMI (34.65 ± 7.43 vs 31.06 ± 5.99 kg/m², *P* < .001), and had slightly advanced Tanner staging (*P* = .002). There was, however, no significant difference found in the Cr (57.46 ± 15.38 vs 54.81 ± 13.43 μ mol/L, *P* = .122), BUN (4.21 ± 1.04 vs 4.46 ± 1.19 mmol/L, *P* = .150) and eGFR (108.94 ± 31.37 vs 110.06 ± 31.87 mL/min/1.73 m², *P* = .808) levels between the two groups. After matching for age, sex, Tanner staging and BMI, no difference remained in the outcomes between the groups.

2. Low HDL and kidney function outcomes

When MUO group with HDL <1.03 mmol/L (<40 mg/dL) (*n* = 84) were compared to MHO (*n* = 105), the groups were comparable in age (12.75 ± 3.30 vs 11.89 ± 3.31 years, *P* = .079) and Tanner staging (*P* = .090). The subgroup of MUO with lower HDL had higher BMI (33.37 ± 6.10 vs 31.06 ± 5.99 kg/m², *P* = .010) and BMI z-score (*P* = .036). BUN was found to be lower (3.88 ± 1.05 vs 4.46 ± 1.2 mmol/L, *P* = .001) in the subgroup vs MHO. There was a trend towards higher Cr (58.34 ± 14.85 vs 54.81 ± 13.43 μ mol/L, *P* = .051) although in the normal range in the subgroup. There was no difference in eGFR (103.80 ± 26.49 vs 110.06 ± 31.87 mL/min/1.73 m², *P* = .151). Also, we found significantly higher TG in those subjects with MUO and low HDL (1.26 ± 0.58 vs 0.91 ± 0.30 mmol/L, *P* < .001). After matching for age, sex, Tanner staging and BMI, the difference in BUN remained with lower BUN (*P* = .009) in the MUO subgroup whereas there was no difference in Cr and eGFR between the groups.

3. Elevated triglycerides and kidney function outcomes

When comparing MUO group with elevated TG level ≥ 1.69 mmol/L (≥ 150 mg/dL) (*n* = 27) and MHO (*n* = 105), the groups were comparable for age (12.27 ± 4.34 vs 11.89 ± 3.31 years, *P* = .624), sex (*P* = .077), Tanner stage (*P* = .957), and BMI (32.18 ± 5.93 vs 31.06 ± 5.99 kg/m², *P* = .387). There were no differences found in BUN (4.25 ± 1.04 vs 4.46 ± 1.2 mmol/L, *P* = .423), Cr (60.11 ± 16.97 vs 54.80 ± 13.43 μ mol/L, *P* = .073) and eGFR (101.10 ± 29.23 vs 110.06 ± 31.87 mL/min/1.73 m², *P* = .188) between the two groups. Also, HDL was lower (*P* < .001) in the MUO with elevated TG subgroup.

4. Blood pressure and kidney function outcomes

The subjects were divided into MUO with SBP or DBP z-score of ≥ 90 th percentile for age, sex and height (*n* = 50) and MHO (*n* = 105). The groups were comparable for age (12.62 ± 3.93 vs 11.89 ± 3.31 years, *P* = .235) and sex (*P* = .799). MUO with high BP had more advanced Tanner stage (*P* = .016), higher BMI (34.67 ± 7.11 vs 31.06 ± 5.99 kg/m², *P* = .001) and BMI z-score (*P* = .026). Although in the normal range, Cr was higher (59.23 ± 15.38 vs 54.82 ± 13.44 μ mol/L, *P* = .049) in the MUO with high BP subgroup. No difference was found in BUN (4.12 ± 1.09 vs 4.46 ± 1.2 mmol/L, *P* = .098) and eGFR (103.63 ± 35.58 vs 110.06 ± 31.87 mL/min/1.73 m², *P* = .260) between the groups. After matching for BMI and Tanner staging, the Cr remained higher (still in the normal range) in the MUO with high BP subgroup (*P* = .047) vs MHO. The subjects

TABLE 1 Characteristics of MUO and MHO groups

	MUO (n = 172), mean $\pm$ SD	MHO (n = 105), mean $\pm$ SD	P-value
Age (y)	12.93 $\pm$ 3.29	11.89 $\pm$ 3.31	.012
Sex, females	42%	56%	.026
Tanner staging			.058
1	16%	30%	
2	12%	9%	
3	11%	12%	
4	23%	23%	
5	36%	25%	
Height (cm)	157.57 $\pm$ 15.15	153.29 $\pm$ 15.85	.026
Height z-score	0.68 $\pm$ 1.15	0.84 $\pm$ 1.24	.289
Weight (kg)	85.66 $\pm$ 27.27	75.36 $\pm$ 26.14	.002
Weight z-score	2.55 $\pm$ 0.56	2.45 $\pm$ 0.60	.147
BMI (kg/m ²)	33.64 $\pm$ 6.74	31.06 $\pm$ 5.99	.001
BMI z-score	2.36 $\pm$ 0.32	2.29 $\pm$ 0.35	.080
SBP (mm Hg)	116.23 $\pm$ 10.59	109.34 $\pm$ 9.20	<.001
SBP z-score	0.66 $\pm$ 0.94	0.16 $\pm$ 0.65	<.001
DBP (mm Hg)	65.87 $\pm$ 7.81	63.51 $\pm$ 6.69	.011
DBP z-score	0.16 $\pm$ 0.68	0.02 $\pm$ 0.61	.102
A1C (%)	5.62 $\pm$ 0.69	5.15 $\pm$ 0.32	<.001
TG (mmol/L)	1.14 $\pm$ 0.52	0.91 $\pm$ 0.31	<.001
HDL (mmol/L)	1.12 $\pm$ 0.28	1.35 $\pm$ 0.25	<.001
BUN (mmol/L)	4.13 $\pm$ 1.09	4.46 $\pm$ 1.20	.023
Creatinine (μ mol/L)	58.36 $\pm$ 14.68	54.82 $\pm$ 13.44	.027
eGFR (mL/min/1.73 m ²)	105.39 $\pm$ 28.99	110.06 $\pm$ 31.87	.211

Abbreviations: BMI, body mass index; BUN, blood urea nitrogen; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; HDL, high-density lipoprotein; MHO, metabolically healthy obesity; MUO, metabolically unhealthy obesity; SBP, systolic blood pressure; TG, triglycerides.

with SBP or DBP z-score of ≥ 90 th percentile for age, sex and height (n = 50) were also compared to the rest of the group (n = 226). There were no differences found in Cr ($P = .212$), BUN ($P = .327$) and eGFR ($P = .380$) between the two groups.

The above subjects were stratified further into those with high DBP (n = 7) vs high SBP (n = 43). The seven subjects with DBP ≥ 90 th percentile for age, sex and height included six of those that had both SBP and DBP ≥ 90 th percentile for age, sex and height. The groups were comparable for age ($P = .387$), sex ($P = .533$), Tanner stage ($P = .386$), absolute height ($P = .787$) and BMI ($P = .521$). SBP ($P = .178$) and SBP z-scores ($P = .076$) were similar in both the groups. The group with the high DBP had a higher Cr, although in the normal range ($P = .017$) with a trend towards higher eGFR ($P = .056$). There was no significant difference in the BUN ($P = .423$) between the two groups.

3.3 | Effect of BMI ≥ 40 kg/m² vs < 40 kg/m² on kidney function outcomes

The study subjects were divided into those with BMI ≥ 40 (n = 39) vs < 40 kg/m² (n = 238) to study the effect of severe obesity on kidney function. The subjects with BMI ≥ 40 kg/m² were older (15.47 ± 2.38

vs 12.06 ± 3.22 years, $P < .001$) and taller (165.19 ± 1.20 vs 154.43 ± 15.63 cm, $P < .001$). The absolute SBP, SBP z-score and DBP were higher ($P < .001$, $P = .021$ and $P = .013$, respectively) in the group with BMI ≥ 40 kg/m² whereas DBP z-score ($P = .306$) was similar between the groups. The higher BMI group had a trend towards higher Cr although still in the normal range (60.99 ± 15.55 vs 56.57 ± 14.05 μ mol/L, $P = .054$). No difference was found in BUN (4.09 ± 0.82 vs 4.29 ± 1.18 mmol/L, $P = .320$) and eGFR (108.81 ± 37.60 vs 106.89 ± 28.82 mL/min/1.73 m², $P = .713$) between the groups.

After the subjects were matched for height, age, sex and Tanner stage, no difference was found in the BUN ($P = .958$), Cr ($P = .975$) and eGFR ($P = .436$) between the groups.

3.4 | Effect of degree of obesity vs metabolic health on glomerular hyperfiltration

Glomerular hyperfiltration was defined by eGFR ≥ 135 mL/min/1.73 m². Glomerular hyperfiltration was found in 9% of MHO and 11% of MUO. Using chi-square tests, there was no difference in the glomerular hyperfiltration distribution between the two groups ($P = .6$). There was also no difference found in the eGFR between subjects with MUO and normal A1C of $\leq 5.6\%$ vs those with A1C $> 5.6\%$

($P = .182$). When all subjects were stratified into BMI z-score ≥ 2.5 vs < 2.5 , glomerular hyperfiltration was seen in 13% and 16%, respectively. The percentages of subjects with glomerular hyperfiltration did not differ ($P = .5$).

4 | DISCUSSION

There are only a few studies examining kidney function in MHO and MUO children and adolescents.¹¹ In our retrospective analysis of a total of 277 children and adolescents with obesity that were categorized based on their metabolic health, we found no significant difference in the eGFR and Cr between the MHO and MUO groups. However, a higher BUN in the MHO group was possibly explained by higher dietary protein intake, not accounted for during the study period. The levels of BUN and less so Cr are affected by dietary protein intake.²⁰ After age, sex and Tanner stage adjusted, there were no differences in all three outcomes. A similar difference in BUN was found when the groups were categorized based on their HDL. Lower BUN and higher TG levels were found in subjects with MUO who had lower HDL suggesting the possibility of higher dietary protein intake in metabolically healthy subjects vs higher fat intake in those who were metabolically unhealthy.²⁰ The trend towards a higher Cr seen in the MUO subjects with lower HDL could be explained by a higher BMI; which disappeared once the groups were matched for their BMI.

Higher Cr level, although in normal range, in our metabolically unhealthy subjects with elevated BP, especially with elevated DBP, suggest the effect of higher BP on kidney function decline. Studies in children also have shown that essential hypertension is an independent risk factor for reduced Cr clearance.²¹ A small rise in systolic and diastolic BPs have been shown to increase Cr levels especially in young African-American men.²² Elevated DBP having a small but significant rise in Cr levels and also a negative effect on eGFR, may be an early indicator of kidney function decline. Children and adolescents with MUO have an unfavourable cardiometabolic risk profile, which may be associated with CKD that can arise in adulthood.

In this study, morbid obesity with BMI ≥ 40 kg/m² also did not appear to significantly affect kidney function. Glomerular hyperfiltration can be an early sign of ORG. The effect of obesity on increases in the plasma volume and activation of renin angiotensin system leading to increased filtration has been described.²³ Even though a relatively high percentage of children in our study were found to have glomerular hyperfiltration, no significant difference was found in subjects with healthy or unhealthy metabolic statuses or even in those with higher degree of obesity. There was also no difference found in the glomerular hyperfiltration rate based on the glycaemic status of those that were metabolically unhealthy.

Use of Cr and BUN as measures of kidney function has limitations because of effects of muscle mass and dietary protein intake.²⁴ eGFR, calculated by the modified Schwartz formula, has been found to underestimate GFR in adolescents with normal kidney function; therefore, may also not be an accurate estimate of the actual GFR.²⁵ Inulin clearance is considered gold standard for the evaluation of GFR.

Cystatin-based GFR has been shown to be a more accurate estimate of GFR and is independent of the effect of age, muscle mass and body composition.^{19,26} Even though inulin clearance and cystatin-based formulas are more accurate and reliable measures of eGFR, these test are expensive and are technically difficult, and thus are not ideal for screening purposes.²⁷ There is a need for other markers to identify early kidney function decline. Several studies have found microalbuminuria as an early indicator for ORG.¹² The detection of microalbuminuria has not been adapted as a standard screening test but could be used for identification of ORG and prevention of ESKD in children with obesity. Due to the retrospective nature of our study, laboratory data to assess for microalbuminuria was not available. Also, kidney function decline associated with obesity may not be identified early during childhood and adolescence and may manifest later in adulthood due to its slow process.

4.1 | Limitations

Our study was retrospective and cross-sectional; thus, the inherent nature of the study could not be corrected for. Since the subjects were not followed over time, the possible progressive effect of obesity on kidney function decline could not be studied.

5 | CONCLUSION

Cr, eGFR and BUN levels did not suggest kidney function decline in children and adolescents with obesity independent of their metabolic health status and the degree of obesity. Higher Cr, although in normal range, occurred in those with abnormal DBP. Nevertheless, the follow-up of these children is prudent and the standardization of screening approach to identify CKD at an early stage to prevent the onset of ORG and progression to ESKD is needed.

CONFLICT OF INTEREST

No conflict of interest was declared.

AUTHOR CONTRIBUTIONS

S.A. is the primary author responsible for the literature search, data collection, interpretation and writing of the manuscript. V.U. is the principal investigator responsible for the conception of the research idea, data analysis and interpretation, reviewing and editing of the manuscript. Co-author V.L.C. is responsible for data interpretation and reviewing of the manuscript. Co-author L.D. is responsible for data collection and reviewing of the manuscript. Co-author L.M.W. is responsible for data collection and reviewing of the manuscript. All authors had final approval of the submitted version.

ETHICS STATEMENT

This study has been approved by the IRB committee at SUNY Downstate Health Sciences University.

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